Assignment ครั้งที่ 6

วิชา 060243104 Object-oriented Programming

ตอนที่ 3

โดย

68-060216-1203-7 นายปภาวิน ธิติชุณหกุล

ห้อง IT 1-RC

เสนอ

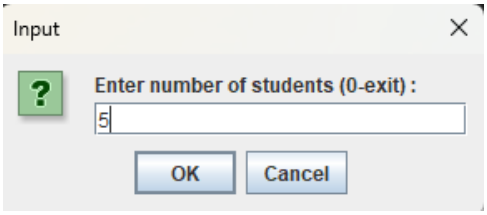
ผู้ช่วยศาสตราจารย์สมชัย เชียงพงศ์พันธุ์

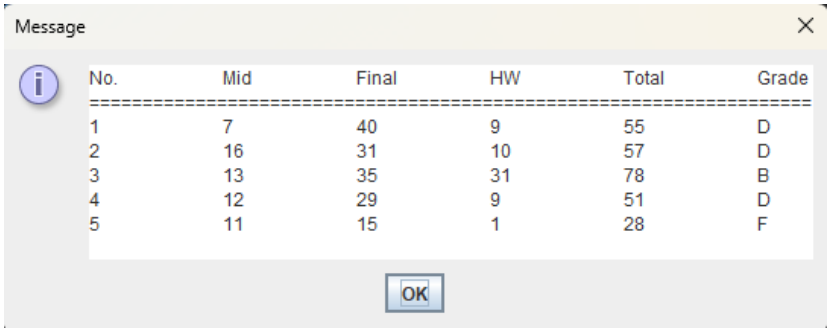
ภาควิชาเทคโนโลยีสารสนเทศ

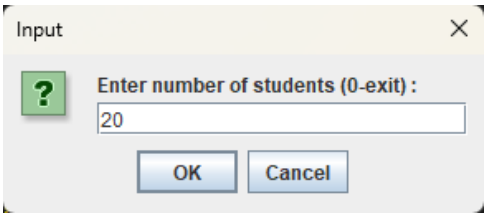
คณะเทคโนโลยีและการจัดการอุตสาหกรรม

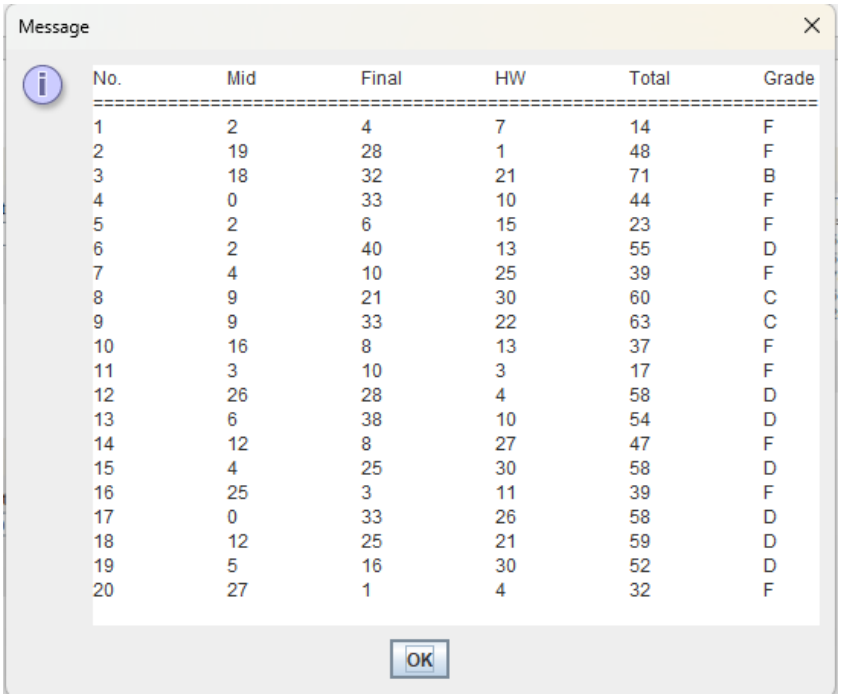
Assignment #036
060243104 Object-oriented Programming

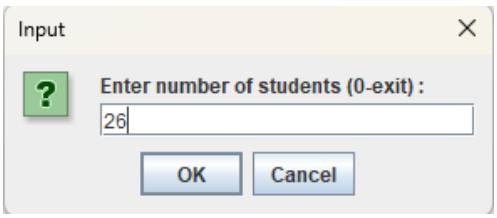
```java
// Id    :  68-060216-1203-7
// Name  :  Mr.Paphavin Thitichunhakun
// Room  :  1 RC
// File Name Assign06_6806021612037.java
import java.text.DecimalFormat;
import java.util.Random;

import javax.swing.JOptionPane;
import javax.swing.JTextArea;

public class Assign06_6806012612037 {
    String output;
    JTextArea textArea;
    Random rnd = new Random();
    Score score;
    String dataScore[];
    int count = 0;
    public static void main(String[] args) throws Exception {
        Assign06_6806012612037 obj = new Assign06_6806012612037();
        obj.mainProgram();
    }

    public void mainProgram() {
        while (true) {
            String input = JOptionPane.showInputDialog("Enter number of students (0-exit) : ");
            if (input == null || input.equals("0")) {JOptionPane.showMessageDialog(null, "Exit Program..."); System.exit(0);}
            else if (input.isEmpty()) { System.exit(0);}
            else if (Integer.parseInt(input) <= 20 && Integer.parseInt(input) > 0) {
                dataScore = new String[(Integer.parseInt(input) * 5)];
                count = 0;
                output = "No.\tMid\tFinal\tHW\tTotal\tGrade\n==================================================================\n";
                for (int i = 0; i < Integer.parseInt(input); i++) {
                    dataScore[count] = String.valueOf(i + 1); genScore();
                }
                report();
            }
            else JOptionPane.showMessageDialog(null, "Number of students not correct.");
        }

    }

    public void genScore() {
        dataScore[count + 1] = String.valueOf(rnd.nextDouble(31)); dataScore[count + 2] = String.valueOf(rnd.nextDouble(41)); dataScore[count + 3] = String.valueOf(rnd.nextDouble(31));
        score = new Score(Double.parseDouble(dataScore[count + 1]) , Double.parseDouble(dataScore[count + 2]), Double.parseDouble(dataScore[count + 3]));
        dataScore[count + 4] = score.toGrade();
        output += dataScore[count] + "\t" + score.toString();
        count += 4;
    }

    public void report() {
        textArea = new JTextArea(output);
        JOptionPane.showMessageDialog(null, textArea);
    }
}

class Score {
    private double midtermExam = 0.0;
    private double finalExam = 0.0;
    private double homework = 0.0;

    public Score() {}

    public Score(double m, double f, double h) {
        midtermExam = m;
        finalExam = f;
        homework = h;
    }

    public double getMidtermExam() {
        return midtermExam;
    }

    public void setMidtermExam(double midtermExam) {
        this.midtermExam = midtermExam;
    }

    public double getFinalExam() {
        return finalExam;
    }

    public void setFinalExam(double finalExam) {
        this.finalExam = finalExam;
    }

    public double getHomework() {
        return homework;
    }

    public void setHomework(double homework) {
        this.homework = homework;
    }

    public double toSum() {
        return midtermExam + finalExam + homework;
    }

    public String toGrade() {
        String grade;
        if (toSum() >= 80) grade = "A";
        else if (toSum() >= 70) grade = "B";
        else if (toSum() >= 60) grade = "C";
        else if (toSum() >= 50) grade = "D";
        else grade = "F";
        return grade;
    }

    public String toString() {
        DecimalFormat twoDigit = new DecimalFormat("#0");
        return twoDigit.format(midtermExam) + "\t" + twoDigit.format(finalExam) + "\t" + twoDigit.format(homework) + "\t" + twoDigit.format(toSum()) + "\t" + toGrade() + "\n";
    }
}
```

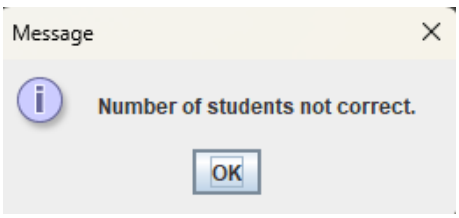

Input
Enter number of students (0-exit) :
5
OK
Cancel

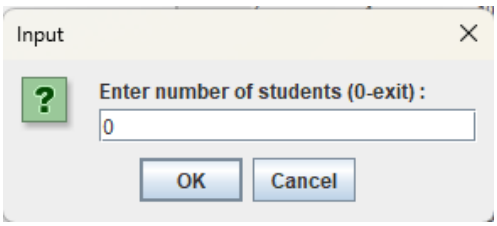

Message
No. Mid Final HW Total Grade
==========
1 7 40 9 55 D
2 16 31 10 57 D
3 13 35 31 78 B
4 12 29 9 51 D
5 11 15 1 28 F
OK

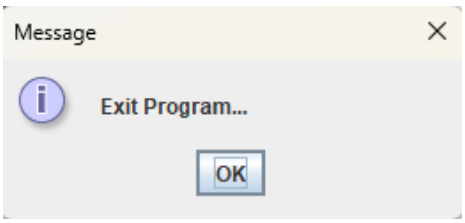

Input
Enter number of students (0-exit) :
20
OK
Cancel


Message
No. Mid Final HW Total Grade
==========
1 2 4 7 14 F
2 19 28 1 48 F
3 18 32 21 71 B
4 0 33 10 44 F
5 2 6 15 23 F
6 2 40 13 55 D
7 4 10 25 39 F
8 9 21 30 60 C
9 9 33 22 63 C
10 16 8 13 37 F
11 3 10 3 17 F
12 26 28 4 58 D
13 6 38 10 54 D
14 12 8 27 47 F
15 4 25 30 58 D
16 25 3 11 39 F
17 0 33 26 58 D
18 12 25 21 59 D
19 5 16 30 52 D
20 27 1 4 32 F
OK


Input
Enter number of students (0-exit) :
26
OK
Cancel


Message
Number of students not correct.
OK


Input
Enter number of students (0-exit) :
0
OK
Cancel


Message
Exit Program...
OK